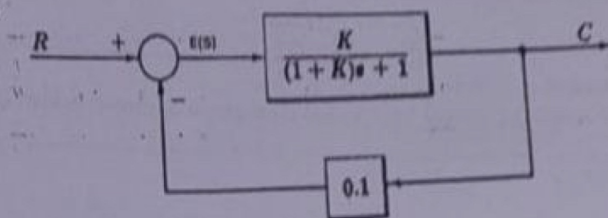




6

For the block diagram shown below,
 $E(S)/R(S) =$

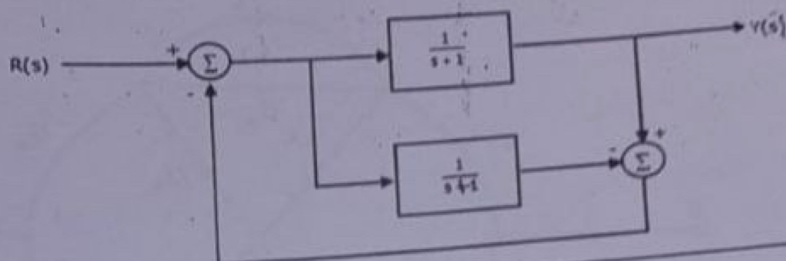
- a) $[(S+1)/((1+K)S+1+0.1K)]$
 b) $[(1+K)S+1/((1+K)S+1+0.1K)]$
 c) $[(1+K)S+1/((K)S+1+0.1K)]$
 d) None



7

The transfer function of the system below is

- a) $\frac{1}{s+1}$ b) $\frac{2}{s+3}$
 c) $\frac{2}{s+1}$ d) None



8

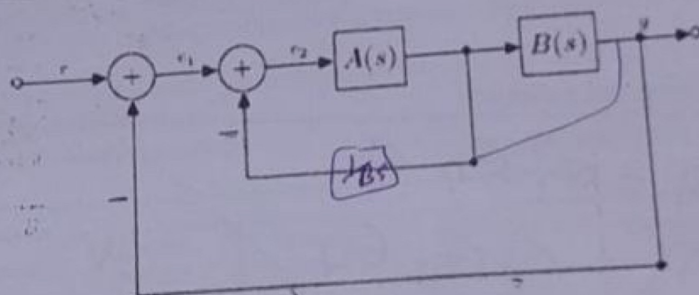
for the following first order system $\frac{C(s)}{R(s)} = \frac{10}{2s+1}$ the Dc.gain is:

- A. 5 B. 6.32 C. 0 D. 10 E. none

9

The transfer function relating
 $C(s)$ and $R(s)$ is

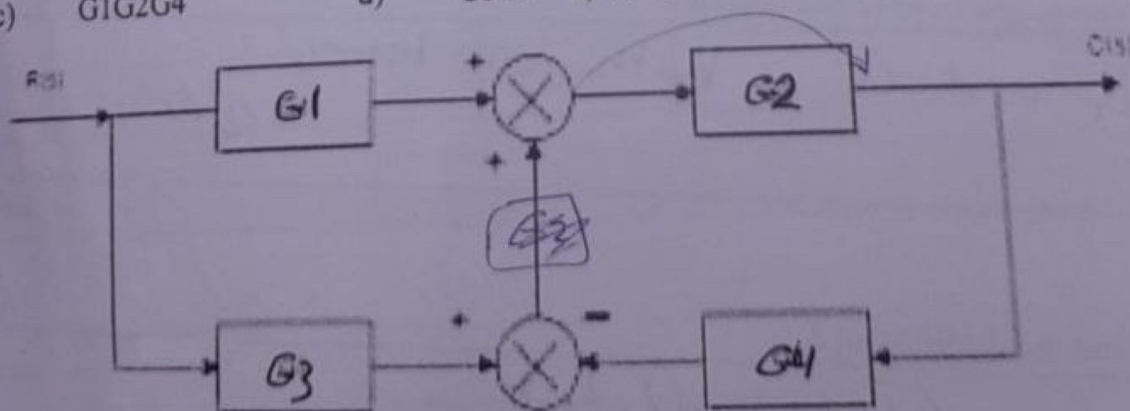
- A. $\frac{B(s)}{1+A(s)+A(s)B(s)}$
 B. $\frac{A(s)B(s)}{1+A(s)+B(s)+A(s)B(s)}$
 C. $\frac{A(s)B(s)}{1+A(s)+A(s)B(s)}$
 D. $\frac{A(s)B(s)}{1-A(s)+A(s)B(s)}$



10

Consider the block diagram shown below. If the transfer function of the system is given by
 $T(s) = G1G2+G2G3/1+X$. Then X is:

- a) $G2G3G4$ b) $G2G4$
 c) $G1G2G4$ d) $G3G4$ e) none



1	2	3	4	5	6	7	8	9	10
a	c	A	A	C	b	C	A	C	b

Select the most accurate answer for each of the following statements:

1. The transfer Function from Fig.1 is:

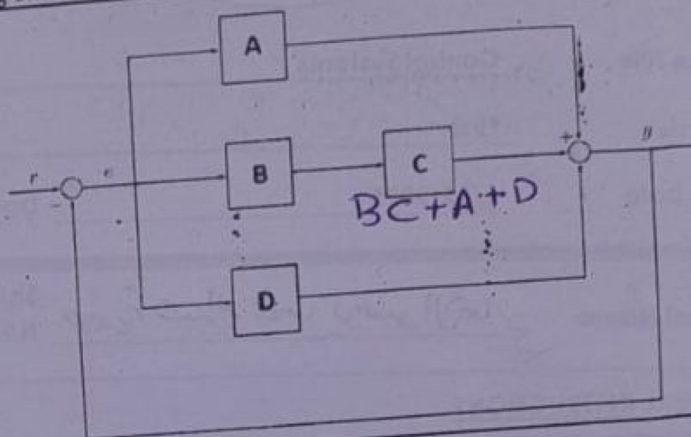
a. $A + BC + D$

b. $\frac{A+BC+D}{1+BC}$

c. $\frac{A+BC}{1+A+BC+D}$

d. $\frac{A+BC}{1+D}$

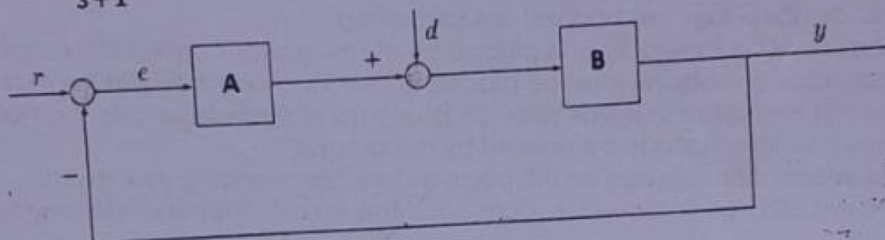
e. none



You are given the Below system diagram. What is the associated transfer function from disturbance to output: y/d

$A = \frac{s+1}{s+2}$

$B = \frac{1}{s+1}$



a. $\frac{1}{s(s+1)}$

b. $\frac{s+1}{s+2}$

c. $\frac{s+1}{(s+2)(s+3)}$

d. $\frac{s+2}{(s+1)(s+3)}$

e) none

Moving the pole of the first order system near to the origin (by making T larger) it will

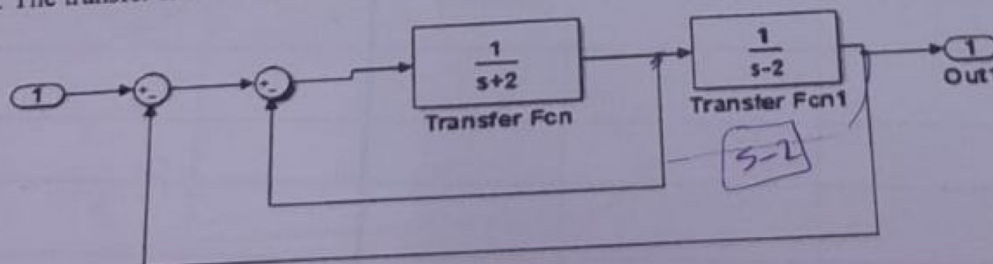
A. make the system slower

B. Make the system faster

D. Decreases the Dc gain

E. none

6. The transfer function of the following system is:



A. $\frac{1}{s^2+s-6}$

B. $\frac{1}{s^2-s-6}$

C. $\frac{1}{s^2+s-5}$

D. $\frac{1}{s^2+s+6}$

E. none

For the following first order system $\frac{C(s)}{R(s)} = \frac{10}{2s+1}$ the input is $r(t)=1$. At $t=0$ the value of $c(t)$ is:

A. 6.32

B. 2

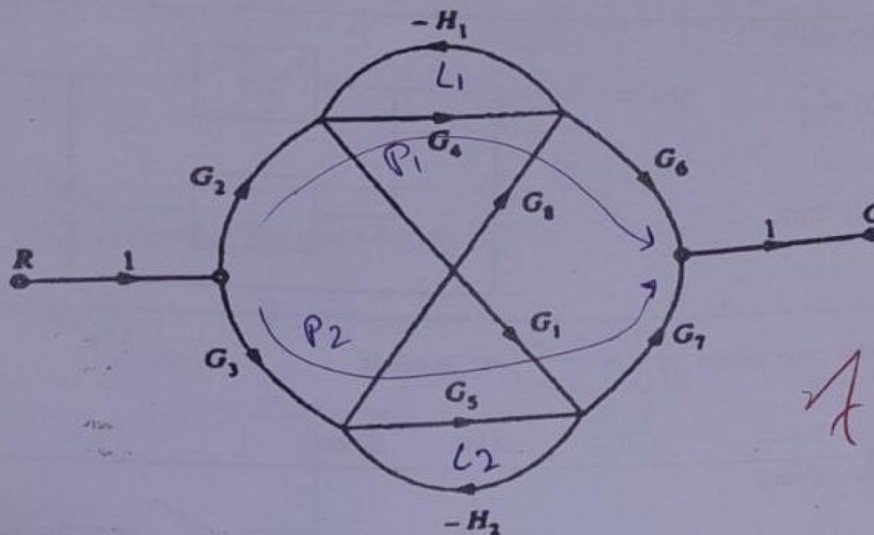
C. 0

D. 4

E. none

Q2 (10 points)

Consider the control system whose signal flow graph is shown below. Determine the system transfer function using Mason's formula



1) Forward paths are:

$$P_1 = G_2 G_4 G_6$$

$$P_2 = G_3 G_5 G_7$$

2) Feedback loops (close loops) are:

$$L_1 = -G_4 H_1$$

$$L_2 = -G_5 H_2$$

3) Non-touch loops (2 at a time) are:

L_1 and L_2 both non-touch

4) Find the determinant Δ

$$\Delta = 1 - (L_1 + L_2) + (L_1 L_2)$$

$$= 1 - (-G_4 H_1 + -G_5 H_2) + (+G_4 H_1 G_5 H_2)$$